

Transport and Curvature Certificates for the Totient Constant

a Lean 4 companion to the Erdős #249 formalisation

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Abstract

Let $S = \sum_{n \geq 1} \varphi(n)/2^n$ and $R_N = \sum_{j \geq 1} \varphi(N+j)/2^j$. The irrationality of S is Erdős Problem 249 and remains open. We formalise exact secondary reductions built from finite differences of the tails R_N : curvature, exponent-transport and prime-jump commutators, and an anchored four-vertex operator at $H, 3H, 5H, 15H$. Each object splits into an integer dyadic window and a rigorously bounded tail, giving a decidable modular certificate for non-integrality. We prove that old Möbius divisor channels are affine and are annihilated by the relevant moments, isolate the new-channel migration at a prime jump, and show that every fixed-precision valuation-unit word admits a centred completion. The finite consumers and no-go theorem are unconditional; every irrationality conclusion remains conditional on an explicitly named unbounded supply. No such supply, and hence no solution of Problem 249, is claimed.

1 Scope and logical boundary

The main exposition [3] develops the direct certificate reduction for S , proves an unconditional denominator exclusion, and records the open boundary. This companion studies finite-difference operators on the same tail sequence. They annihilate affine contributions from divisor channels already present at an lcm height, but the remaining new channels must still be shown to produce non-integral values at unboundedly many heights. Thus every endpoint below has the form

$$\text{unbounded supply of finite certificates} \implies S \notin \mathbb{Q},$$

with the antecedent open. A certificate at one height is finite evidence, not evidence for the cofinal quantifier. The fixed-precision no-go theorem is a boundary on one local strategy, not a global impossibility result.

The Lambert-series background is classical. The open problem is recorded in the Erdős–Graham collection and maintained catalogue [1, 2]. We make no priority claim for finite differences or affine moment cancellation as general techniques. The contribution here is their precise integration into checked identities, finite consumers, conditional sockets, and an obstruction theorem.

2 Dyadic windows

For a finite integer combination $X(H) = \sum_i c_i R_{m_i H}$, truncate each tail at depth L . Multiplication by 2^L gives

$$2^L X(H) = A(H, L) + E(H, L),$$

where $A(H, L) \in \mathbb{Z}$ is computable. If $|E(H, L)| \leq B(H, L)$, then $X(H) \notin \mathbb{Z}$ whenever

$$B(H, L) < A(H, L) \bmod 2^L < 2^L - B(H, L).$$

The reusable soundness theorem is `notMem_int_of_window_remainder_bound` [Lean](#). `Lean` checks the exact decomposition, radius, and concrete modular arithmetic; an external analytic or arithmetic argument must provide unbounded instances.

3 Curvature on the lcm cone

Set

$$Q_H = R_{4H} - 3R_{2H} + 2R_H.$$

Its depth- L window $C(H, L)$ satisfies

$$2^L Q_H = C(H, L) + E_C(H, L), \quad |E_C(H, L)| \leq 6H + 3L + 3.$$

The exact split and bound are `two_pow_mul_curvature_eq_window_add_shifted` [Lean](#) and `abs_curvatureShiftedTail_le` [Lean](#). The central residue criterion therefore proves $Q_H \notin \mathbb{Z}$ by `curvature_notMem_int_of_sharpCurvatureCert` [Lean](#). An unbounded supply implies irrationality through `irrational_totientSeries_of_sharpCurvatureSupply` [Lean](#), but that supply is not proved.

The window evolves by $C(H, L+1) = 2C(H, L) + d(H, L+1)$, checked in `curvatureWindow_succ` [Lean](#). Exact phase-certificate and common-carrier formulas expose possible routes to a supply theorem without asserting one.

4 Exponent transport and affine cancellation

If $p \mid H$, multiplication by p adds no squarefree divisor support:

$$d \mid pH \iff d \mid H \quad (d \text{ squarefree}).$$

This is `squarefree_dvd_mul_iff_of_dvd` [Lean](#). The exact residue kernel is affine, not merely periodic:

$$K_d(N) = N a_d(N) + b_d(N),$$

where both a_d and b_d are periodic modulo d . The decomposition is `transportResidueKernel_eq_affineState` [Lean](#). Keeping both blocks prevents an intercept-only matrix calculation from being misread as global non-cancellation. The exact $H \mapsto 3H$ two-block transfer is `residueTransportDefect_three_eq_affineState` [Lean](#).

Every old channel $d \mid H$ is killed by the first-order defect `residueTransportDefect_eq_zero_of_dvd` [Lean](#). The endpoint `irrational_totient_series_of_exponentOnlyThreeTransportSupply` [Lean](#) is conditional on a named fixed-three tower supply.

More generally, coefficients satisfying $\sum_i c_i = \sum_i c_i m_i = 0$ annihilate every old affine channel. This is `oldChannel_affine_moment_annihilation` [Lean](#). For $q(X, Y) = XY - 3X - 2Y + 4$, the vanishing $q(1, 1) = q(3, 5) = 0$ gives

$$K_d(15H) - 3K_d(3H) - 2K_d(5H) + 4K_d(H) = 0 \quad (d \mid H),$$

checked by `joint35_oldChannel_zero` [Lean](#). The associated window radius is $19H + 5L + 5$; its finite consumer is `joint35Cone_notMem_int_of_cert` [Lean](#). The conditional endpoint is `irrational_totient_series_of_anchoredJoint35ConeSupply` [Lean](#).

5 Prime jumps and three-transport boundaries

When $p \nmid H$, a jump from H to pH creates new squarefree support. The residue Hecke defect splits into a post-jump channel and an explicit migration correction in `residueHeckeDefect_eq_postJump_add_migration` [Lean](#). For $d \mid H$ the total old-channel defect is zero [Lean](#); the $(d, H, p) = (5, 12, 5)$ calculation records the two cancelling terms.

Set

$$J(H, p) = (R_{2pH} - R_{pH}) - p(R_{2H} - R_H).$$

Its exact remainder radius is $3pH + (p + 1)(L + 2)$, proved with the window split by `primeJumpTailCommutator_eq_window_add_remainder` [Lean](#). The finite consumer is `primeJumpTailCommutator_notMem_int_of_sharpKill` [Lean](#). Lean verifies $(H, p, L) = (12, 5, 15)$ in `primeJumpSharpKill_twelve_five` [Lean](#); this is one finite deposit, not an unbounded sequence.

The balanced vertices $H, 2H, 3H, 6H$ give a related radius $9H + 4L + 4$. The consumer is `threeTransport_notMem_int_of_cert` [Lean](#); the irrationality theorem `irrational_totientSeries_of_roughnessBoundaryLatticeSupply` [Lean](#) is conditional. The concrete check $(H, L) = (60, 12)$ is `sharpThreeTransportCert_60_12` [Lean](#).

6 A first-harmonic interface

For the window discrepancy $A(h, N, L)$ define

$$W_1(h, N, L) = \cos \left(2\pi \frac{A(h, N, L) \bmod 2^L}{2^L} \right).$$

Assume $N \in [X, 2X)$ and $16(2X + h + L + 2) \leq 2^L$. If every N failed the central-residue certificate, each phase would lie within $\pi/8$ of an integer turn, so each cosine would exceed $9/10$. Hence

$$\sum_{X \leq N < 2X} W_1(h, N, L) \leq \frac{9}{10}X$$

forces a certificate in the block. This implication is `exists_certifiedKill_of_first_harmonic_gap` [Lean](#). No cofinal exponential-sum estimate establishing its hypothesis is proved.

7 Fixed local precision cannot finish the argument

A finite valuation-unit symbol records a bounded amount of local transport data. For every positive precision u , every finite compatible word admits a prefix-locked centred completion. Thus no finite word over this fixed local alphabet excludes all centred endpoints. The exact statement is `fixedPrecisionTropicalNoGo` [Lean](#). This does not say curvature certificates are rare. A proof may use precision growing with scale, global correlations, or a harmonic gap. It rules out only the inference that a bounded local signature alone forces escape.

8 Statement matrix and open boundary

Layer	Checked statement	Status
Curvature	exact split, radius, recurrence, consumer	unconditional
Exponent transport	stable support, affine state, cancellation	unconditional
Joint (3, 5) transport	moment annihilation and consumer	unconditional
Prime jump	migration, sharp window, (12, 5, 15)	finite
Three transport	sharp window and (60, 12)	finite
First harmonic	cosine gap forces a certificate	conditional input
Tropical signature	fixed precision has a completion	no-go
Irrationality of S	follows from a named unbounded supply	supply open

The development proves no cofinal supply of curvature, prime-jump, joint, or three-transport certificates, and no cofinal first-harmonic gap. It therefore does not prove Erdős #249.

References

- [1] P. Erdős and R. L. Graham, *Old and New Problems and Results in Combinatorial Number Theory*, Monographies de l'Enseignement Mathématique 28 (1980), 61–62, [scan](#).
- [2] T. Bloom (ed.), *Erdős Problems*, entry 249, erdosproblems.com/249, accessed July 2026.
- [3] W. Cook, *Erdős Problems 249 and 257 in Lean 4: formalised results, certificate reductions, and open questions*, companion exposition in this release.